14. Answer : option(a)

For $[MA_4B_2]^{n+}$ complexes-geometrical isomerism is possible

$$\left.\begin{matrix}[Co(NH_3)_4Br_2]Cl \\ [Co(NH_3)_4BrCl]Br\end{matrix}\right\} \rightarrow \text{ionisation isomers}$$

15. Answer : option(d)

Option (a) and (b) – geometrical isomerism is possible

Option (c) – ionization isomerism is possible

Option (d) – no possibility to show either constitutional isomerism or stereo isomerism

16. Answer : option(c)

(a) Fe^{2+} (b) Fe^{3+} (c) Fe^{0}

17. Answer : option(d)

$[Fe(en)_3]^{2+}(PO_4^{3-})$

18. Answer : option(c)

(a) Zn^{2+} $(d^{10} \Rightarrow \text{diamagnetic})$

(b) Co^{3+} $(d^6 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 0}$; diamagnetic $)$

(c) Ni^{2+} $(d^8 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 2}$; paramagnetic $)$

(d) $[Ni(CN)_4]^{2-}$ $(dsp^2$; square planar, diamagnetic$)$

19. Answer : option(c)

$[Co(NH_3)_3(Cl)_3]$

20. Answer : option(d)

V^{2+} $(t_{2g}^{\ 3}, e_g^{\ 0}$; CFSE $= 3 \times (-0.4)\Delta_0 = -1.2\Delta_0)$

Ti^{2+} $(t_{2g}^{\ 2}, e_g^{\ 0}$; CFSE $= 2 \times (-0.4)\Delta_0 = -0.8\Delta_0)$

Statements given in option (a) ,(b), and (c) are wrong.

The current statements are

(a) since, the crystal field stabilization is more in octahedral field , octahedral complexes are more stable than square planar complexes.

(b) $[FeF_6]^{4-}$ – (Fe^{2+} – d^6 High Spin – $t_{2g}^{\ 4}, e_g^{\ 2}$; CFSE $= 4 \times (-0.4) + (0.6) \times 2 + P$)

$[Fe(CN)_6]^{4-}$ – (Fe^{2+} – d^6 Low Spin – $t_{2g}^{\ 6}, e_g^{\ 0}$; CFSE $= 6 \times (-0.4) + 3P$)

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